12. 'Static' keyword

In java, static keyword can be used with→

1. variable
2. method
3. block
4. Inner class

Static variable

- When a variable is declared as *static*, then a single copy of the variable is created and shared among all objects at the class level.
- Static variables are, essentially, global variables.
- All instances [objects] of the class share the same static variable.
- **static block and static variables are executed in the order they are present in a program.[line by line]**
- In two class communication, static variables can be accessed directly using class name.
  Here, consider *Animal* as a class name and *name* is a static field in java class.

  ```
  System.out.println(Animal.name);
  ```

Static Methods

- When a method is declared with the *static* keyword, it is known as the static method.
- The most common example of a static method is the *main( )* method.
- Methods declared as static have several restrictions:
  - They can only directly call other static methods.
  - They can only directly access static data.
  - They cannot refer to this or super in any way.

Static class

- A class can be made **static** only if it is a nested class.
- We cannot declare a top-level class with a static modifier but can declare nested classes as static. Such types of classes are called Nested static classes.
- Nested static class doesn't need a reference of Outer class.
- In this case, a static class cannot access non-static members of the Outer class.

All the codes and practices regarding static keyword are completed in sessions.

When to use static keyword?

with Variable-

- when shared Data Across All Instances is required.
- When you define a constant that shouldn't change and is common to all instances.

with method-

- If the method doesn't use instance variables or methods.
- Utility or helper methods.(e.g. sqrt method in Math class)
- Factory method for connection with other services.

When not to use static keyword?

- When behavior/data should differ across objects.
- When using inheritance or polymorphism.
- When working with instance-specific data.

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